

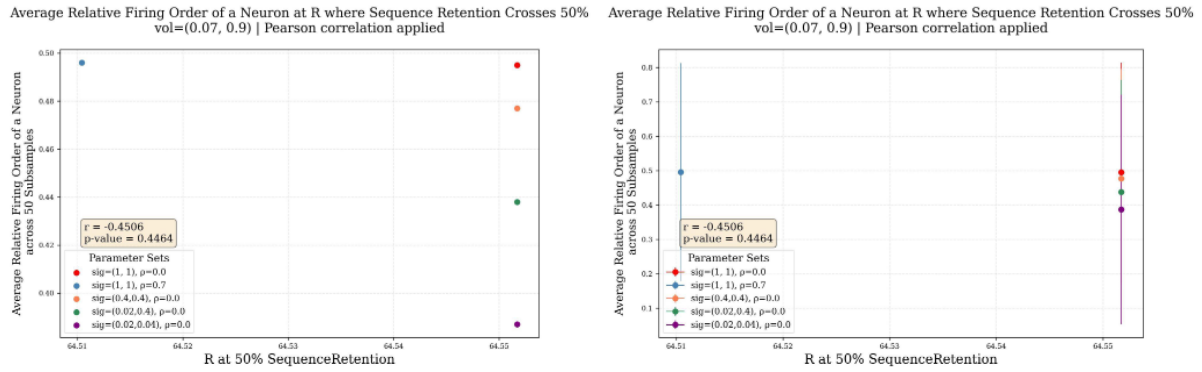


Figure 1. Relative firing order of a selected neuron averaged across different R values for 50 random subsamples against the R values at which 50% of sequences are retained for each parameter set. Pearson's correlation applied.

Relative firing order: In each sequence, neurons are indexed starting from 0 to 1. (The first neuron is 0 and the last neuron is 1). The location of a neuron is then computed relatively.

A high averaged relative order indicates a neuron's firing positing across sequences vary highly. The higher the sigma range, the higher relative order. Changing the sigma while keeping the volume constant did not change sequence retention rate. When sigma is correlated, there is a slight increase in sequence retention rate. Introducing 0.7 correlation shifted R at 50% from 64.55 to 64.51. Although there is observed to be a low correlation between averaged relative firing order of neurons and 50% sequence retention rate, that correlation coefficient is not statistically significant.

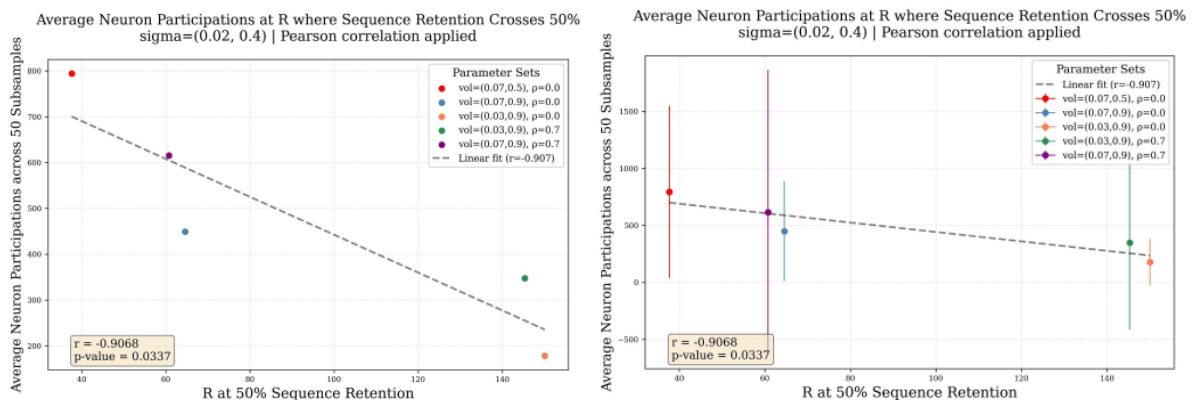


Figure 2. The number of sequences each neuron participates in is averaged across different R values for 50 random subsamples against the R values at which 50% of sequences are retained for each parameter set. Pearson's correlation applied and plotted as a fitting line.

There is observed to be a strong negative correlation between average number of neuron participations and sequence retention rate. The correlation is statistically significant ($p < 0.05$). Correlating volumes increases the average neuron participation as well as the rate of sequence retention. Although this increase in average was previously observed to be due to having a few outlier neurons which are active in the majority of the sequences.